

Multi table – Query

students :

student_id	student_name
1	Alice
2	Bob
3	Charlie

courses :

course_id	student_id	course_name
101	1	Math
102	1	English
103	2	Science

students :	
student_id	student_name

courses :		
course_id	student_id	course_name

Get a list of students and the courses they are enrolled in.

```
SELECT students.student_name, courses.course_name
```

```
FROM students,courses
```

```
Where students.student_id = courses.student_id;
```

students :

student_id	student_name
1	Alice
2	Bob
3	Charlie

courses :

course_id	student_id	course_name
101	1	Math
102	1	English
103	2	Science

student_name	course_name
Alice	Math
Alice	English
Bob	Science

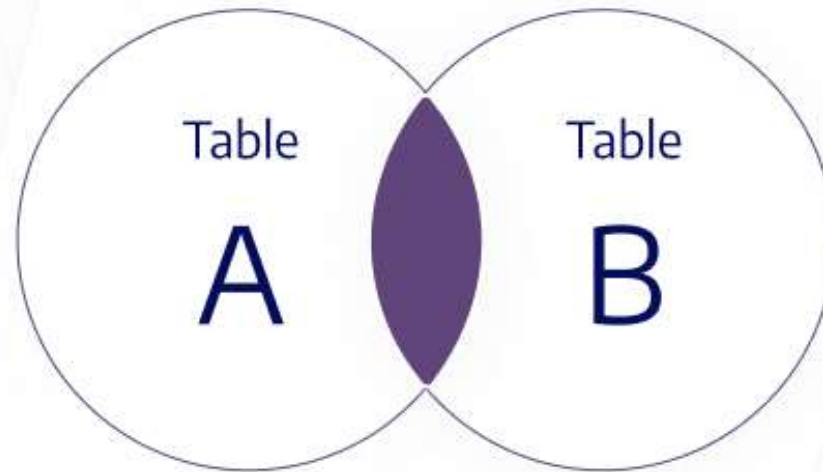
Inner Join

ID	
1	A
2	B
3	C
6	D

Table
A

ID	
5	L
2	M
3	N
4	O

Table
B



2	B	M
3	C	N

first_name	last_name	salary	department_id
William	Shakespeare	1500	50
Emily	Dickinson	2700	50
John	Donne	3800	90
Oscar	Wilde	4200	90

department_id	department_name	region_id
50	IT	700
90	HR	800



first_name	department_name
William	IT
Emily	IT
John	HR
Oscar	HR

Inner Join

students :	
student_id	student_name

courses :		
course_id	student_id	course_name

Get a list of students and the courses they are enrolled in.

```
SELECT students.student_name, courses.course_name  
  
FROM students  
  
INNER JOIN courses ON students.student_id = courses.student_id;
```

students :

student_id

student_name

courses :

course_id

student_id

course_name

Get a list of students who are enrolled in the "Math" course.

```
SELECT s.student_name, c.course_name
```

```
FROM students s, courses c
```

```
WHERE s.student_id = c.student_id and c.course_name = 'Math';
```


students :

student_id

student_name

courses :

course_id

student_id

course_name

Get a list of students who are enrolled in any course
(without repeating their names).

```
SELECT distinct (students.student_name)
FROM students, courses
WHERE students.student_id = courses.student_id;
```

students :

student_id

student_name

courses :

course_id

student_id

course_name

Count how many courses each student is enrolled in.

```
SELECT students.student_name, COUNT(*) AS course_count
FROM students, courses
Where students.student_id = courses.student_id
GROUP BY students.student_name;
```

Note: The above query does not provide the count for the students who has not registered for any courses

students :

student_id	student_name
1	Alice
2	Bob
3	Charlie

courses :

course_id	student_id	course_name
101	1	Math
102	1	English
103	2	Science

students :

student_id

student_name

courses :

course_id

student_id

course_name

Count how many courses each student is enrolled in.

```
SELECT s.student_name, COUNT(c.course_id) AS course_count
FROM students s LEFT JOIN courses c
ON s.student_id = c.student_id
GROUP BY s.student_name;
```

Note: The above query will provide the count for the students who has not registered for any courses also

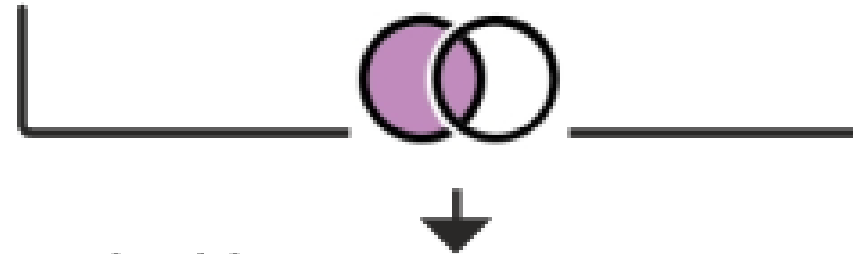
Left Table

Date	CountryID	Units
1/1/2024	1	40
1/2/2024	1	25
1/3/2024	3	30
1/4/2024	4	35

Right Table

ID	Country
1	USA
2	Canada
3	Panama

Left Join



Merged Table

Date	CountryID	Units	Country
1/1/2024	1	40	USA
1/2/2024	1	25	USA
1/3/2024	3	30	Panama
1/4/2024	4	35	<i>null</i>

salesman

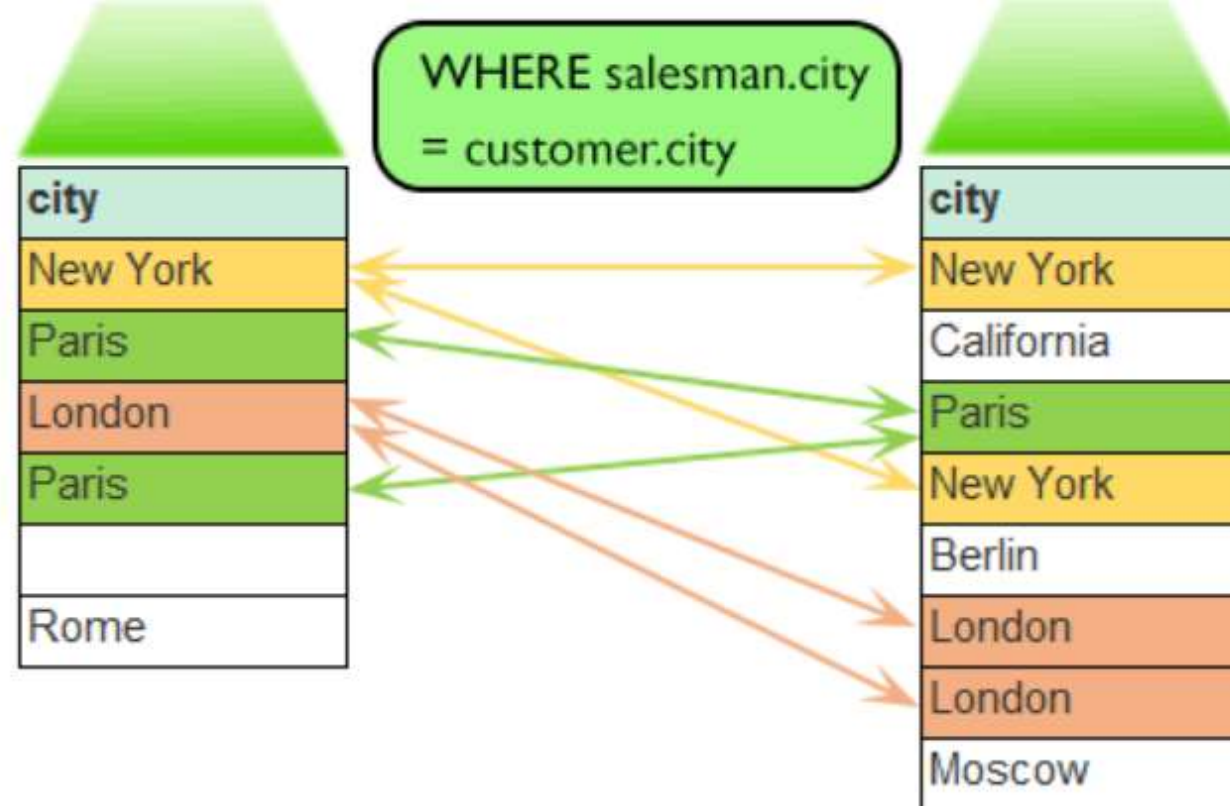
salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14
5007	Paul Adam	Rome	0.13
5003	Lauson Hen	San Jose	0.12

customer

customer_id	cust_name	city	grade	salesman_id
3002	Nick Rimando	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	California	200	5002
3008	Julian Green	London	300	5002
3004	Fabian Johnson	Paris	300	5006
3009	Geoff Cameron	Berlin	100	5003
3003	Jozy Altidor	Moscow	200	5007

write a SQL query to find the salespeople and customers who live in the same city. Return customer name, salesperson name and salesperson city.

```
SELECT customer.cust_name,  
salesman.name, salesman.city  
FROM salesman, customer  
WHERE salesman.city = customer.city;
```



cust_name	name	city
Brad Davis	James Hoog	New York
Nick Rimando	James Hoog	New York
Fabian Johnson	Nail Knite	Paris
Brad Guzan	Pit Alex	London
Julian Green	Pit Alex	London
Fabian Johnson	Mc Lyon	Paris

write a SQL query to locate all the customers and the salesperson who works for them. Return customer name, and salesperson name

```
SELECT customer.cust_name, salesman.name  
FROM customer,salesman  
WHERE salesman.salesman_id = customer.salesman_id;
```

salesman

salesman_id	name	city	commission
5001	James Hoog	New York	0.15
5002	Nail Knite	Paris	0.13
5005	Pit Alex	London	0.11
5006	Mc Lyon	Paris	0.14
5007	Paul Adam	Rome	0.13
5003	Lauson Hen	San Jose	0.12

orders

ord_no	purch_amt	ord_date	customer_id	salesman_id
70001	150.5	2012-10-05	3005	5002
70009	270.65	2012-09-10	3001	5005
70002	65.26	2012-10-05	3002	5001
70004	110.5	2012-08-17	3009	5003
70007	948.5	2012-09-10	3005	5002
70005	2400.6	2012-07-27	3007	5001
70008	5760	2012-09-10	3002	5001

customer

customer_id	cust_name	city	grade	salesman_id
3002	Nick Rimando	New York	100	5001
3007	Brad Davis	New York	200	5001
3005	Graham Zusi	California	200	5002
3008	Julian Green	London	300	5002
3004	Fabian Johnson	Paris	300	5006
3009	Geoff Cameron	Berlin	100	5003
3003	Jozy Altidor	Moscow	200	5007

write a SQL query to find those salespeople who generated orders for their customers but are not located in the same city. Return ord_no, cust_name, customer_id (orders table), salesman_id(order table)

```
SELECT ord_no, cust_name, orders.customer_id, orders.salesman_id
FROM salesman, customer, orders
WHERE customer.city <> salesman.city
AND orders.customer_id = customer.customer_id
AND orders.salesman_id = salesman.salesman_id;
```

write a SQL query to locate the orders made by customers. Return order number, customer name.

```
SELECT orders.ord_no, customer.cust_name  
FROM orders, customer  
WHERE orders.customer_id = customer.customer_id;
```

write a SQL query to find those customers who are served by a salesperson and the salesperson earns commission in the range of 12% to 14% (Begin and end values are included.). Return cust_name AS "Customer", city AS "City"

```
SELECT customer.cust_name AS "Customer",  
customer.city AS "City",  
salesman.name AS "Salesman",  
salesman.commission  
FROM customer,salesman  
WHERE customer.salesman_id = salesman.salesman_id  
AND salesman.commission  
BETWEEN .12 AND .14;
```

write a SQL query to find all orders executed by the salesperson and ordered by the customer whose grade is greater than or equal to 200. Compute purch_amt*commission as "Commission". Return customer name, commission as "Commission%" and Commission.

```
SELECT ord_no, cust_name, commission AS "Commission%",  
purch_amt*commission AS "Commission"  
FROM salesman,orders,customer  
WHERE orders.customer_id = customer.customer_id  
AND orders.salesman_id = salesman.salesman_id  
AND customer.grade>=200;
```


write a SQL query to find those customers who placed orders on October 5, 2012. Return customer_id, cust_name, city, grade, salesman_id, ord_no, purch_amt, ord_date, customer_id and salesman_id.

```
SELECT *  
FROM customer a,orders b  
WHERE a.customer_id=b.customer_id  
AND b.ord_date='2012-10-05';
```